Course Code: CS2201 **Name of the Programme: Bsc (Hons)** **Name of the Course: Database Management Systems** **Semester: II**		
Course Credit	**No. of Hours Per Week (L + T + P)**	**Total No. of Teaching Hours**
3	**1:1:2**	**60**
Course Objectives:		
a) Understand the Core Concepts of Database Management Systems		
b) Apply Data Modeling Techniques to Design Efficient Databases		
c) Develop Proficiency in SQL for Data Manipulation and Querying		
d) Demonstrate practical skills in implementing and managing databases while analyzing their role in real-world applications.		

Course outline (Syllabus of the course)

Module - 1: Introduction to Database Systems	**No. of Hours**
	6 Hrs
Definition and Importance of Databases, Database system architecture, Characteristics of DBMS vs. File Systems, Advantages & Disadvantages of DBMS, Types of Databases (Hierarchical, Network, Relational), Data Independence, Database Users and Administrators, Three-tier Architecture (External, Conceptual, Internal Levels) Data Models: Hierarchical, Network, Relational, Object-Oriented.	

Module - 2: Data Modelling Using ER-Model and Constraints	**No. of Hours**
	6 Hrs
Entity Types, Entity Sets, Attributes, and Keys, Relationship Types, Introduction to ER Diagrams, Components of ER Diagrams, Drawing ER Diagrams, Specialization, Generalization & Aggregation, Mapping ER Diagrams to Relational Schema, Integrity constraints, entity integrity, referential integrity, Keys constraints, Domain constraints.	

Module - 3 : SQL Queries	**No. of Hours**
	9 Hrs
Database language and interfaces - DDL, DML, DCL and TCL, Characteristics of SQL, advantages of SQL. SQL data type and literals, Types of SQL commands, SQL operators and their procedure, Tables, views and indexes, Queries and subqueries, Aggregate functions. Insert, update and delete operations, Joins, Unions, Intersection, Minus, Triggers and Procedures in SQL.	

Module - 4 : Normalization	No. of Hours
	4 Hrs
Introduction to Normalization, Data Anomalies, Functional Dependencies, Types of Normal Forms, 1 Normal Form, 2 Normal Form, 3 Normal Form, BCNF(Boyce & Codd Normal Form.).	

Module - 5 : NOSQL	No. of Hours
	5 Hrs
Introduction to NoSQL, Types of NoSQL Databases, Advantages & Disadvantages of NoSQL, Basics of MongoDB, Querying in NoSQL (MongoDB)	

Course Outcomes :
a) Explore core database concepts, including architecture and transactions in DBMS.
b) Design and implement database applications using ER diagrams to solve real-world problems.
c) Apply query processing techniques and normalization to optimize database performance and eliminate redundancy.
d) Understand the fundamentals and applications of NoSQL databases for handling non-relational data.

Text Books:	
1	Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database system concepts* (7th ed.). McGraw-Hill. ISBN 978-9390727506
2	Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson. ISBN 978-9332582705
3	Sadalage, P. J., & Fowler, M. (2012). *NoSQL distilled: A brief guide to the emerging world of polyglot persistence*. Addison-Wesley Professional. ISBN 978-0321826626

Reference Books:	
1	Ramakrishnan, R., & Gehrke, J. (2002). *Database management systems*. McGraw-Hill Higher Education. ISBN 0072465638
2	Vaswani, V. (2004). *MySQL: The complete reference*. Osborne/McGraw-Hill. ISBN 0072224770
3	Sullivan, D. (2015). *NoSQL for mere mortals*. Addison-Wesley. ISBN 0134023218
4	Hows, D., Membrey, P., & Plugge, E. (2014). *MongoDB basics*. Apress. ISBN 148420896X

Tutorials	
1	Design a University Management System using DDL commands (CREATE, ALTER, and DROP) in MySQL.
2	Perform DML (Data Manipulation Language) operations on a database designed for an online shopping system. This activity covers INSERT, UPDATE, DELETE, and SELECT statements.
3	Apply SQL functions (String, Numeric, Date, and Aggregate functions) using a simple School Database scenario.
4	Implement Date and Conversion Functions in a School Database.
5	Implementing Arithmetic, Logical, and Comparison Operators in SQL.
6	Analyze product inventory and customer orders using special operators and set operations.
7	Implement the different types of join for the given scenario.
8	Implement GROUP BY, HAVING, and ORDER BY in MySQL.
9	Implement a Stored Procedure in MySQL.
10	Implement Triggers in MySQL for INSERT, UPDATE, and DELETE.

List of Experiments	
1	Implement DDL Commands in SQL: Demonstrate the use of DDL commands with suitable examples, including CREATE TABLE, ALTER TABLE, and DROP TABLE.
2	Implement DML Commands in SQL: Implement DML commands with suitable examples, including INSERT, UPDATE, and DELETE.
3	Implement SQL Functions: Implement various types of functions in SQL with examples, including Number Functions, Aggregate Functions, Character Functions.
4	Implement SQL Functions: Conversion Functions, and Date Functions.
5	Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.
6	Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples Special Operators, and Set Operations(UNION)
7	Implement different types of Joins: Implement various types of joins in SQL, including Inner Join, Outer Join, and Natural Join, with suitable examples.
8	Implement Grouping and Ordering: Study and implement the GROUP BY and HAVING clauses, ORDER BY clause.
9.	Implement Procedure.(Basic procedure, procedure for Insert, Update and Delete)
10	Implement Triggers(validation before and after insert, delete and update)

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Lesson Plan

Name of the Programme:	Bsc.(Hons.)	
Title of the Course:	Database Management Systems	
Course Code	CS2201	
Semester:	II	
Name(s) of the Course Instructor(s)	Prof. Soumik Banerjee, Prof. Dr. Lokanayaki K	
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
3	1:1 : 2	60

Session	**Module No.**	**Topic**	**Pedagogy / Activities**	**Pre-reading / Reference**
1	1	DBMS vs File System	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
2		Tutorial: Database Architecture & Components	Group Discussion on Storage Components	Silberschatz, Korth, Sudarshan
3		Lab: DDL Commands – CREATE, ALTER	Hands-on Activity	SQL Reference
4		Lab: DDL Commands – DROP TABLE	Hands-on Activity	SQL Reference
5	1	Database System Architecture	PPT, Discussion	Silberschatz, Korth, Sudarshan

6		Tutorial: Schema and Instances	Discussion on Types of Schemas	Silberschatz, Korth, Sudarshan
7		Lab: DML – INSERT, UPDATE	Hands-on Activity	SQL Reference
8		Lab: DML – DELETE, SELECT	Hands-on Activity	SQL Reference
9	1	Data Independence	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
10		Tutorial: DB Users and Administrators	Role-play Activity	Silberschatz, Korth, Sudarshan
11		Lab: SQL Functions – Number, Aggregate	Hands-on Activity	SQL Reference
12		Lab: SQL Functions – Character, Date	Hands-on Activity	SQL Reference
13	2	ER Model Concepts and Notations	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
14		Tutorial: Draw ER Diagram (real-world)	Diagram Drawing Activity	Silberschatz, Korth, Sudarshan
15		Lab: Mapping ER to Tables (Part 1)	Hands-on Activity	SQL Reference

16		Lab: Mapping ER to Tables (Part 2)	Hands-on Activity	SQL Reference
17	2	Integrity Constraints – Entity, Referential	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
18		Tutorial: Integrity Constraints	Apply Constraints Practically	Silberschatz, Korth, Sudarshan
19		Lab: SQL Operators – Arithmetic, Logical	Hands-on Activity	SQL Reference
20		Lab: SQL Operators – Special, Set	Hands-on Activity	SQL Reference
21	2	Key and Domain Constraints	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
22		Tutorial: Key Types & Uses	Case Study: Key Constraints	Silberschatz, Korth, Sudarshan
23		Lab: SQL Joins – Inner, Outer	Hands-on Activity	SQL Reference
24		Lab: SQL Joins – Natural Join	Hands-on Activity	SQL Reference
25	3	Introduction to SQL & Data Types	PPT, Group Discussion	Silberschatz, Korth, Sudarshan

26		Tutorial: SQL Queries Basics	Practice on SELECT, WHERE	HackerRank / SQL Platform
27		Lab: GROUP BY, ORDER BY	Hands-on Activity	SQL Reference
28		Lab: HAVING Clause	Hands-on Activity	SQL Reference
29	3	SQL Views and Indexes	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
30		Tutorial: Using Views in Queries	Discussion + Examples	SQL Reference
31		Lab: SQL Subqueries	Hands-on Activity	SQL Reference
32		Lab: SQL Nested Queries	Hands-on Activity	SQL Reference
33	3	Aggregate Functions in SQL	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
34		Tutorial: Practical Queries (Mid-level)	Lab-Based Tutorial	SQL Reference
35		Lab: Procedures in SQL	Hands-on Activity	MySQL Docs
36		Lab: Triggers in SQL	Hands-on Activity	MySQL Docs
37	3	Functions in SQL	PPT, Group Discussion	Silberschatz, Korth, Sudarshan

38		Tutorial: Practical Queries (Mid-level)	Lab-Based Tutorial	SQL Reference
39		Implementation of Procedure-Insert	Case Studies and Real world scenarios	SQL Docs
40		Implementation of Procedure-update and delete	Case Studies and Real world scenarios	SQL Docs
41	3	Review of SQL queries	Example based activity and Group discussions	SQL reference
42	4	Tutorial: FD & Anomalies	Example-Based Learning	Silberschatz, Korth, Sudarshan
43		Lab: Convert to 1NF, 2NF	Hands-on Activity	SQL Reference
44		Lab: Convert to 3NF, BCNF	Hands-on Activity	SQL Reference
45	4	Introduction to Normalization	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
46		Tutorial Normalization of tables	Hands on Activity	SQL reference
47		Lab: Convert to 1NF, 2NF	Hands-on Activity	SQL Reference
48		Lab: Convert to 3NF, BCNF	Hands-on Activity	SQL Reference

49	4	Normalization and types of Normalization	PPT, Group Discussion	Silberschatz, Korth, Sudarshan
50	5	Tutorial: Relational vs NoSQL	Compare Models	MongoDB Docs
51		Implementation of Triggers- before	Case Studies and Real world scenarios	SQL Docs
52		Implementation of Triggers- after	Case Studies and Real world scenarios	SQL Docs
53	5	MongoDB Schema & Use Cases	PPT, Case Studies	MongoDB Docs
54		Tutorials on procedures	Case Studies and Real world scenarios	SQL Docs
55		Lab Review of SQL Queries	Hands on	Lab 1 to 10
56		Lab Review of SQL Queries	Hands on	Lab 1 to 10
57	5	Advantages and Disadvantages of NOSQL	PPT, Case Studies	MongoDB Docs
58		Tutorial: implementation of all SQL Queries	Scenario based examples	SQL Docs, PPT
59		Lab Review of SQL Queries	Hands on	Lab 1 to 10

60		Lab Review of SQL Queries	Hands on	Lab 1 to 10

Assessment and Evaluation

Continuous Internal Evaluation (CIE) - 70 Marks	
Components of CIE	**Weightage of each component (Marks)**
CIE 1 - ● MCQ ● ER-Diagram (Pen and Paper)	 10 10
CIE 2 - Pen and Paper	25
CIE 3 - SQL queries execution (Lab CIE) / Viva - Record(github)	15+5 5

Semester End Exams (SEE) - 30 Marks	
Mode of Exam	Practical
Weightage of the exam	30%

Course Outcomes- Programme Outcomes Matrix

PO / **CO**		**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	Explore the database concepts, architecture and transaction for DBMS.	3	2	2	-	1	-	1	2	-	-	-	-
CO2	Apply query processing techniques to automate the real time problems of databases..	3	3	2	1	1	-	-	2	-	-	-	-

CO3	Design a database application to solve real world problems using ER Diagram.	3	3	3	1	1	-	-	2	-	-	-	-
CO4	Identify and solve the redundancy problem in database tables using normalization.	3	3	3	-	1	-	-	3	-	2	-	-

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE Components		**Rubrics for Assessment**
CIE-1	MCQ	Multiple choice questions shall be provided, each question carrying 1 mark. for 20 MCQs. Time limit 15 min. no negative marks
CIE-1	Pen and paper ER-Diagram	Scenario shall be provided on which ER-Diagram has to be drawn in paper, valuated on basic of notations and cardinality relationships

Total Marks		**25**
CIE Components		**Rubrics for Assessment**
CIE-2	Theory	The test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

<table>
<tr><th colspan="2">Total Marks</th><th>25</th></tr>
<tr><th colspan="2">CIE Components</th><th>Rubrics for Assessment</th></tr>
<tr><td rowspan="3">CIE-3</td><td>Lab Practical</td><td>The test will be conducted for 15 marks and evaluated based on the rubrics</td></tr>
<tr><td>Record</td><td>
<table>
<tr><th>Criteria</th><th>Marks</th><th>Description</th></tr>
<tr><td>Program Accuracy</td><td>2</td><td>Correct logic, SQL syntax, expected output shown</td></tr>
<tr><td>Record Completeness</td><td>1</td><td>All assigned programs are neatly written and present</td></tr>
<tr><td>Input/Output Format</td><td>1</td><td>Proper sample input and output screenshots/prints</td></tr>
<tr><td>Timely Submission</td><td>1</td><td>Submitted on or before the deadline</td></tr>
</table>
</td></tr>
<tr><td>Viva</td><td>
<table>
<tr><th>Criteria</th><th>Marks</th><th>Description</th></tr>
<tr><td>Conceptual Understanding</td><td>2</td><td>Clear understanding of DBMS fundamentals (e.g., data
models, keys, normalization,
SQL basics)</td></tr>
<tr><td>Application and Practical skills</td><td>1</td><td>Ability to apply concepts to queries, schema design, and
ER diagrams</td></tr>
<tr><td>Accuracy of responses</td><td>1</td><td>Correct and confident
answers to questions asked</td></tr>
<tr><td>Communication and clarity</td><td>1</td><td>Clarity in explanation, use of technical terms, confidence</td></tr>
</table>
</td></tr>
</table>

Table-1: CIE-3 Lab Practical & Viva (Total 25 Marks)

No.	Criteria	Marks	Excellent (Full Marks)	Good (~75%)	Satisfactory (~50–60%)	Not Satisfactory (~≤25%)
1	SQL Queries Execution	15	Logic & Syntax (8M): All queries correct, perfect syntax, exact output.I/O Format (3M): Clear, labeled inputs & outputs.Timely Submission (2M): On/before deadline.Application Skills (2M): Applies concepts effectively to queries, schema, ER diagrams.	Minor syntax/logic errors, mostly correct output; minor I/O issues; small delay; little guidance needed.	Several errors; partial output; unclear I/O; late ≤1 day; significant guidance needed.	Major errors; incorrect/missing output; missing I/O; late >1 day; unable to apply concepts.
2	Viva	5	Conceptual Understanding (3M): Thorough DBMS knowledge (models, keys, normalization, SQL basics).Accuracy & Communication (2M): Correct answers, clear terms, confident delivery.	Mostly clear, minor gaps; small help needed; minor hesitation.	Basic understanding; needs guidance; some correct answers; lacks clarity.	Cannot explain concepts; mostly wrong answers; unclear speech.
3	Record (GitHub)	5	All assigned programs uploaded, neat, complete, and well-documented.	1–2 programs/issues missing; minor formatting/documentation issues.	Several programs missing; incomplete documentation.	Largely incomplete or missing.

Table-2: Semester End Examination (Practical)

Sl. No	Assessment Criteria	Marks	Excellent (Full Marks)	Good (75% Marks)	Satisfactory (50% Marks)	Not Satisfactory (0–25% Marks)

1	Understanding the Problem & Requirements	5	Query design and chosen tables/entities fully match given problem statement; all relationships and constraints correctly identified and implemented.	Mostly correct entities and relationships; 1–2 minor gaps in requirements.	Partial mapping of problem to queries; some entities/relationships missing or incorrect.	Poor mapping; wrong or irrelevant entities; fails to address core requirements.
2	Query Design & Logic	5	Logic is correct and efficient; uses appropriate SQL constructs.	Mostly correct logic with minor issues.	Partially correct logic; uses only basic constructs.	Incorrect or illogical query design.
3	Query Syntax Accuracy	4	Syntax accurate; no errors.	Minor syntax errors, easily corrected.	Multiple syntax issues; partially correct.	Frequent syntax errors; query does not run.
4	Query Execution & Output	5	Executes correctly; output fully matches expected result.	Executes with minor mismatches.	Runs but output partially correct.	Query fails or produces incorrect output.
5	Optimization & Efficiency	3	Optimized, clean query.	Minor inefficiencies.	Acceptable but not optimized.	Poorly written or redundant query.
6	Presentation & Style	3	Well-formatted, indented, and readable query.	Mostly neat, some inconsistencies.	Poor formatting but readable.	Messy, hard to read.
7	Viva Voce	5	Confident, correct answers; clear understanding.	Most answers correct.	Struggles with some concepts; vague explanations.	Unable to explain queries or concepts.

Soumik Banerjee

Signature of the Course Faculty